Lecture Notes: Scheduling, Introduction to Scheduling

Math F113X Math and Society

Goal:

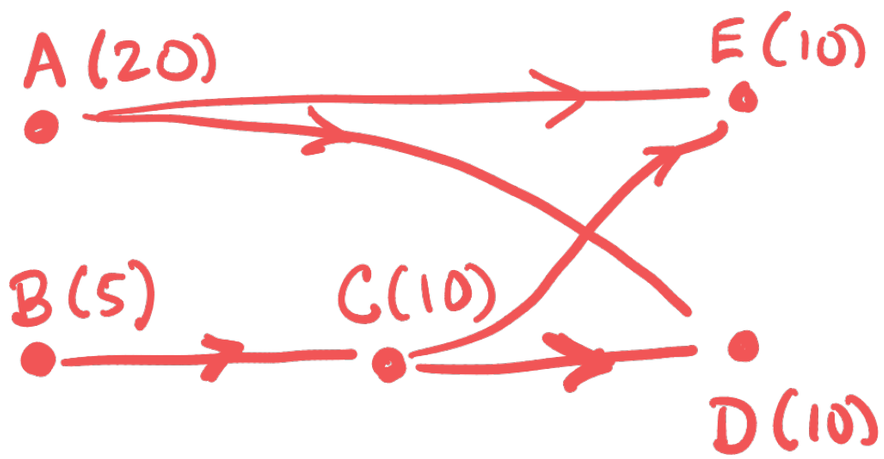
Learn about the following terminology: schedule, digraph, processors, finishing time, optimal finishing time, optimal schedule, idle time, critical path, critical time.

Real World Example: Carnegie Mellon Road Construction Plan

https://www.cmu.edu/cee/projects/PMbook/09_Construction_Planning.html/#9.3%20Defining%20Work%20Tasks

Example: Fixing a Flat Bike Tire

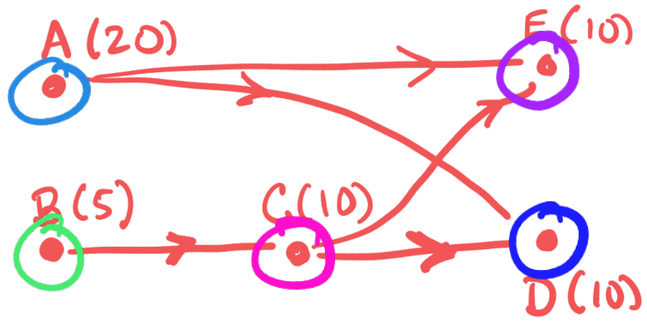
label	task	time	dependence
A	buy a new tube and patch kit	20 minutes	—
B	find tools	5 minutes	—
C	remove tire and old tube	10 minutes	B
D	insert new tube, reattach tire	10 minutes	A B C
E	repair old tube	10 minutes	A B C



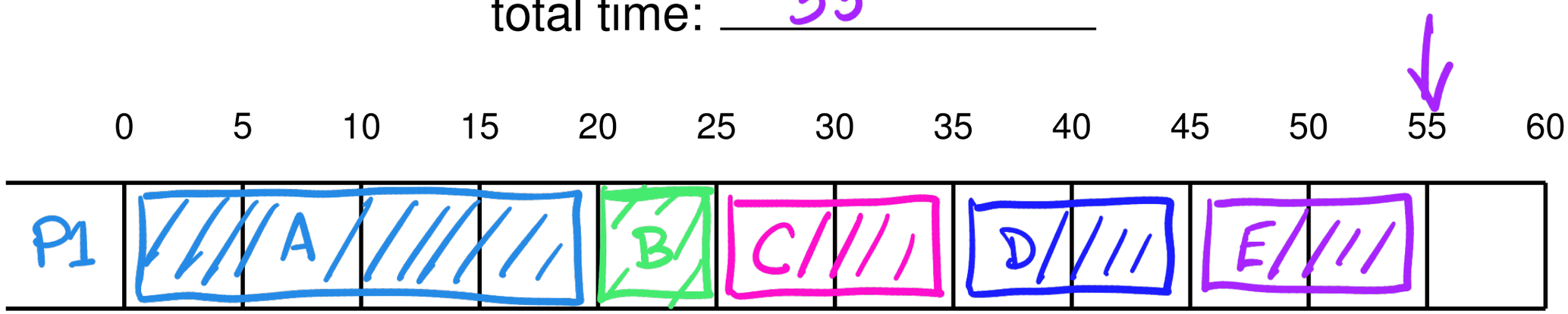
← digraph
= directed graph

Schedule with one processor

label	task	time	dependence
A	buy a new tube and patch kit	20 minutes	—
B	find tools	5 minutes	—
C	remove tire and old tube	10 minutes	B
D	insert new tube, reattach tire	10 minutes	ABC
E	repair old tube	10 minutes	ABC



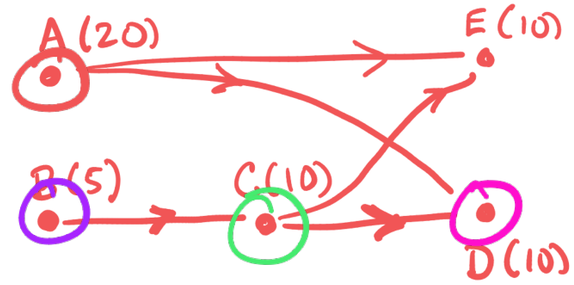
total time: 55



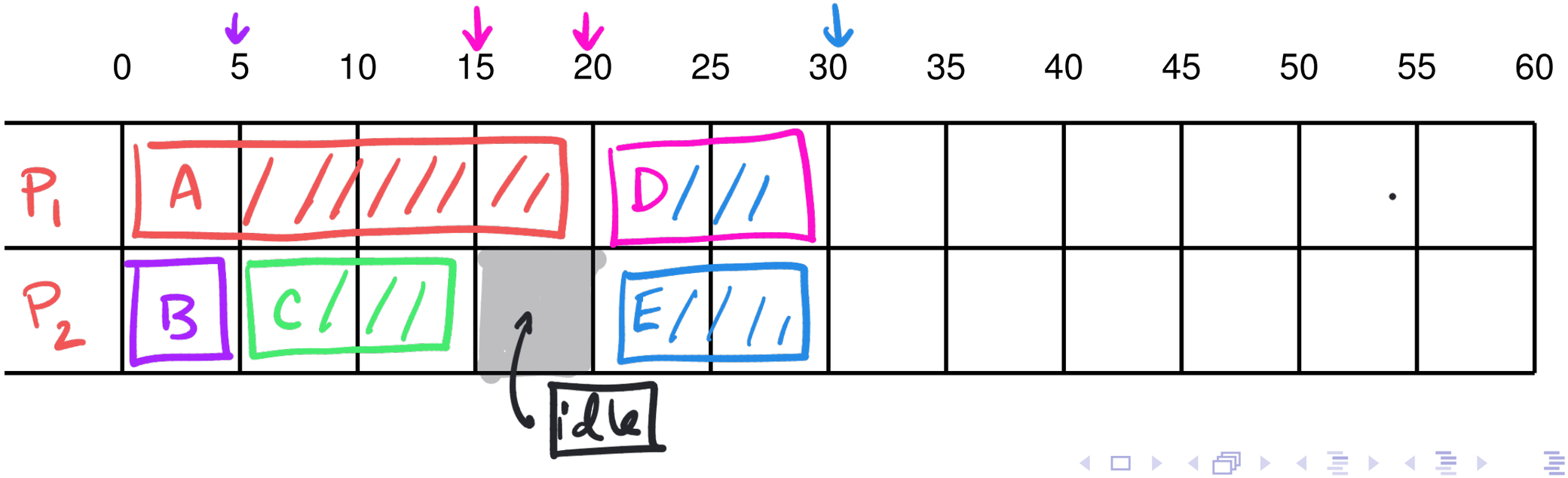
Schedule with two processors

task	A	B	C	D	E
time	20	5	10	10	10
dependence	—	—	B	ABC	ABC

total time: 30



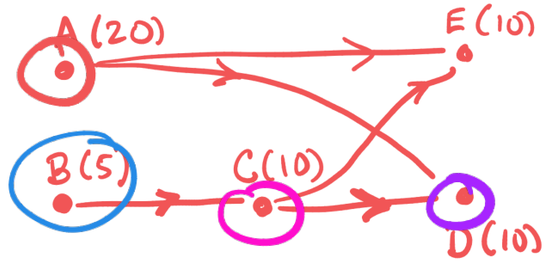
time	0	5	15	20	30	
ready	A B	C	—	D E		
done	—	B	C	A	D, E	done!



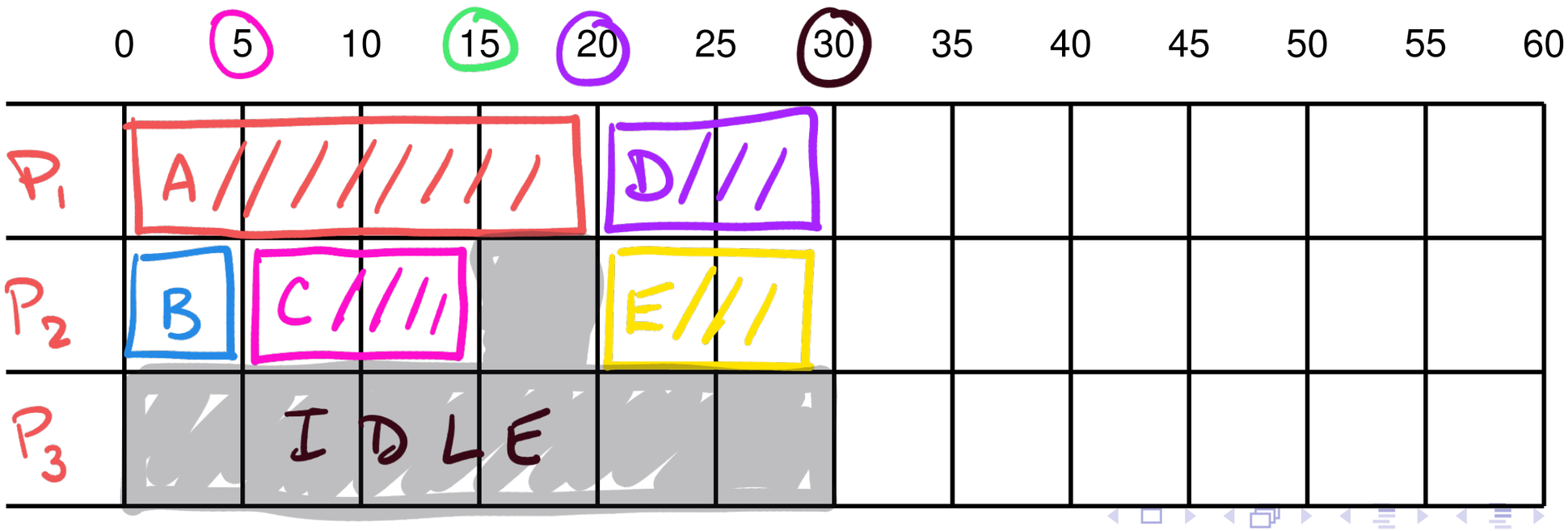
Schedule with three processors

task	A	B	C	D	E
time	20	5	10	10	10
dependence	—	—	B	ABC	ABC

total time: 30



time	0	5	15	20	30
ready	A B	C		DE	
done	—	B	C	A	DE ← done



Terminology

(a) **schedule**

The ordering of all tasks, accounting for dependencies

(b) **digraph**

Model of the schedule

Vertices = tasks (labeled with time)

directed edges $V \longrightarrow W$

You can't complete W until you've completed V

Terminology

(c) **processors**

people / computers / teams completing tasks

(d) **finishing time**

How long it takes to finish the project / complete all tasks

Terminology

(e) **optimal finishing time and optimal schedule**

Shortest possible finishing time and the corresponding schedule assigning tasks to processors

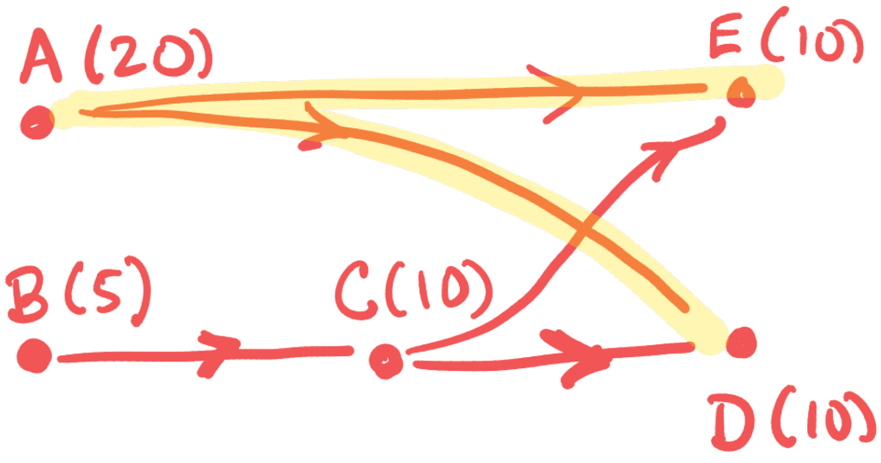
(f) **idle time**

time when processors are not completing tasks

Terminology

(g) **critical path**

Length of longest path according to the sum of the vertex weights



(h) **critical time**

The sum of the vertex weights on the critical path

Paths:

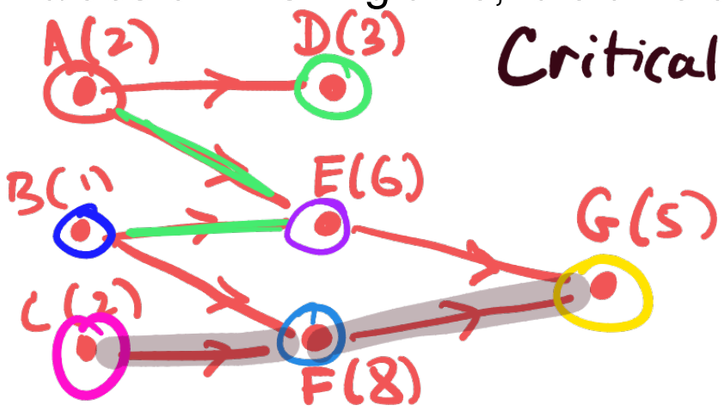
A → E	30
A → D	30
B → C → E	25
B → C → D	25

* optimal time must be at least as long as the critical time!

General Example

label/task	time	dependence
A	2	
B	1	
C	2	
D	3	A
E	6	A, B
F	8	B, C
G	5	E, F

Make a valid schedule with TWO processors. Determine values of finishing time, idle time and critical time.



Critical Path C → F → G time = 15

Finishing time = 15

Optimal!

6 units of idle time.

